

SYDNEY TECHNICAL HIGH SCHOOL



HIGHER SCHOOL CERTIFICATE ASSESSMENT TASK 3

JUNE 2013

Mathematics

General Instructions

- Working time - 70 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in questions 6 to 13
- Start each question on a new page
- A table of standard integrals is provided at the back of the paper

Total marks - 55

Section 1 - 5 marks

Attempt Questions 1 – 5.
Allow about 7 minutes for this section.

Section 2 - 50 marks

Attempt Questions 6 – 11.
Allow about 63 minutes for this section.

Name : _____

Teacher : _____

Section 1

5 marks

Attempt Questions 1 – 5

Allow about 7 minutes for this section

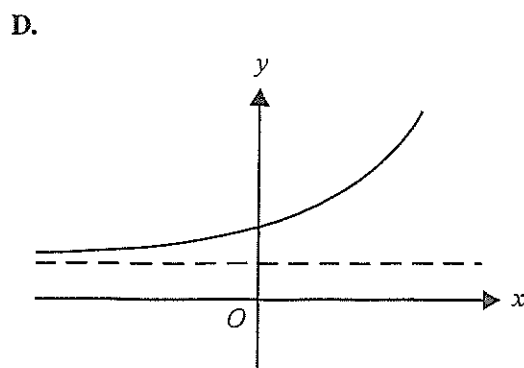
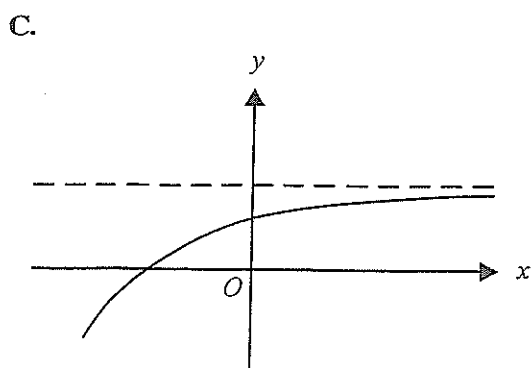
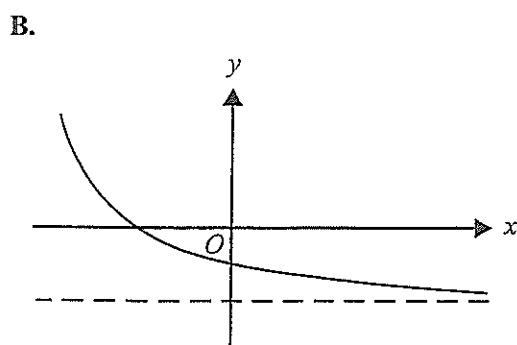
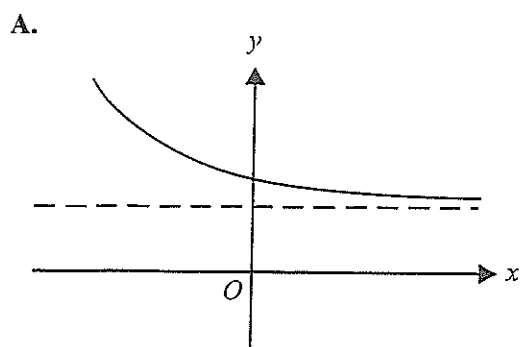
Use the multiple-choice answer sheet in your answer booklet for Questions 1 – 5.

Do not remove the multiple-choice answer sheet from your answer booklet.

1. What is the period of the function $y = 5 - 3 \cos 2x$?

- A) 3
- B) 5
- C) 4π
- D) π

2. If k is a negative real number and P is a positive real number, which one of the following is most likely to be the graph of the function with equation $y = e^{kx} + P$?



3. If $\int_0^a \sec^2 2x \, dx = \frac{1}{2}$, then a is equal to

A. $\frac{\pi}{4}$

B. $\frac{\pi}{8}$

C. $\frac{\pi}{12}$

D. $\frac{\pi}{2}$

4. If $g(t) = e^{-t} - 1$ then $g'(0)$ equals

A. $-e$

B. -2

C. -1

D. 0

5. If $\int_1^3 f(x) \, dx = 5$ then $\int_1^3 (2f(x) - 3) \, dx$ is equal to

A. 4

B. 5

C. 7

D. 10

Section 2

50 marks

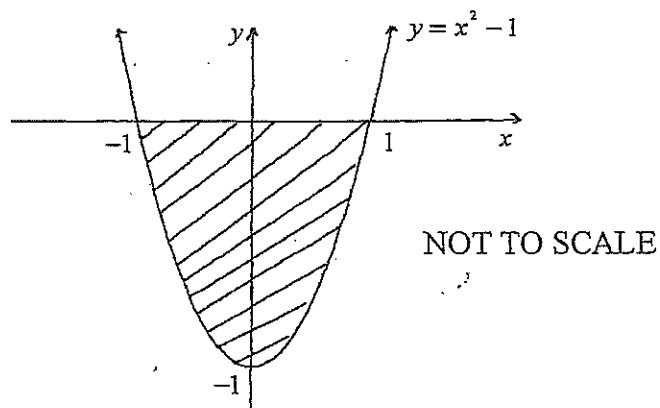
Attempt Questions 6 – 10

Allow about 63 minutes for this section

Start each question on a new page

Question 6 (10 marks)

- a) Evaluate $e^{-2.8}$ giving your answer correct to 2 significant figures. 2
- b) Find the exact value of $\cos \frac{5\pi}{6}$. 1
- c) Sketch the graph of $y = 1 - \cos x$ for $0 \leq x \leq 2\pi$. 2
- d) Evaluate $\int_0^{\frac{\pi}{6}} \sin 2x \, dx$ 2
- e) The area bounded by $y = x^2 - 1$ and the x -axis is rotated about the y -axis. Find the volume of the solid of revolution formed. 3



Question 7 (10 marks) (Start a new page)

a) Solve $\sqrt{3} \tan \theta - 1 = 0$ for $0 \leq \theta \leq 2\pi$. 2

b) Differentiate with respect to x .

i) $\tan 3x + \sin x$ 2

ii) $(x + 2)e^{2x}$ 2

c) Consider the function $f(x) = \frac{x^2}{x+4}$

i) Use the trapezoidal rule with 4 function values to approximate 3

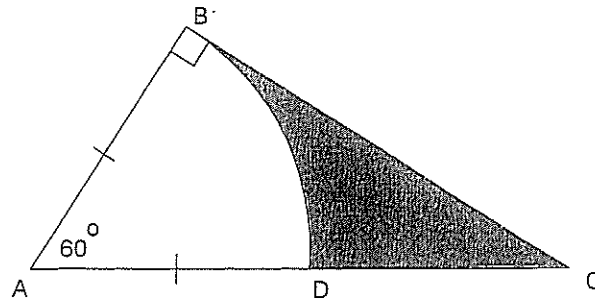
$\int_1^7 f(x) dx$, giving your answer correct to 1 decimal place.

ii) For all values of x , between 1 and 7, $f(x) > 0$, $f'(x) > 0$
and $f''(x) > 0$.

Use this information to decide whether the approximation found 1
in part i) is an over-estimate or an under-estimate of the true value
of the integral. Give a brief reason.

Question 8 (10 marks) (Start a new page)

a)



In the diagram above, angle $B = 90^\circ$, angle $A = 60^\circ$ and $AB = AD = 10 \text{ m}$.

BD is an arc of the circle with centre A .

- | | | |
|-----|---|---|
| i) | Calculate the exact length of the arc BD . | 1 |
| ii) | Calculate the shaded area in exact form. | 3 |
| | | |
| b) | The area bounded by $y = x^2$ and $y = 4$ is rotated about the x -axis to form a solid. Find the volume of the solid. | 3 |
| | | |
| c) | Find the equation of the tangent to $y = e^{4x} + x$ at the point where $x = 0$. | 3 |

Question 9 (10 marks) (Start a new page)

- a) i) Draw a neat sketch of the curve 2

$$y = 3 \sin \frac{x}{2} \text{ for } -2\pi \leq x \leq 2\pi ,$$

showing clearly all the important features.

- ii) Draw on your diagram a line, clearly labelled, which can be 1
used to solve the following equation :

$$3 \sin \frac{x}{2} - x - 1 = 0$$

- iii) Determine the number of solutions the equation 1

$$3 \sin \frac{x}{2} - x - 1 = 0 \text{ has over the domain } -2\pi \leq x \leq 2\pi .$$

- b) i) Show that $\frac{d^2}{dx^2} (e^x \sin x) = 2e^x \cos x$ 2

- ii) Hence find $\int e^x \cos x \, dx$ 2

- c) For what values of k does $y = 3e^{kx}$ satisfy the equation $\frac{d^2 y}{dx^2} - 9y = 0$? 2

Question 10 (10 marks) (Start a new page)

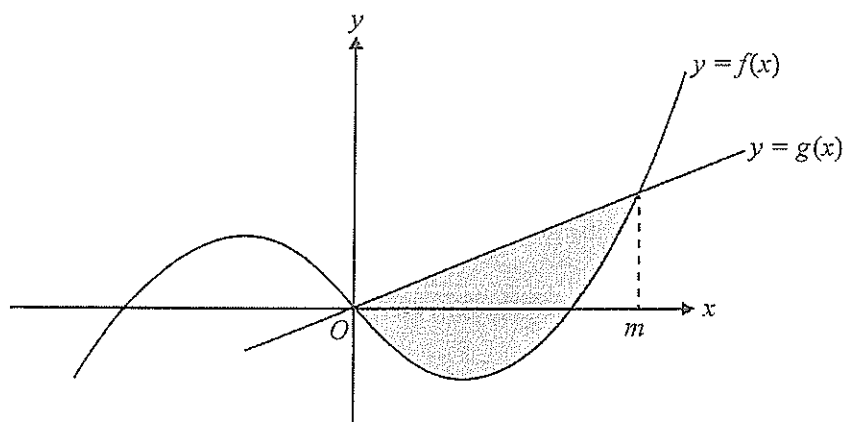
a) A function $f(x)$ is defined by $f(x) = \frac{e^{-x}}{x}$.

i) Differentiate $f(x)$ with respect to x . 2

ii) Find the coordinates of any stationary points of the graph of $y = f(x)$, and determine their nature. 2

iii) Sketch the graph of $y = f(x)$, showing all important features. 2
(Not inflexion points)

b)



Parts of the graphs of the functions $f(x) = x^3 - ax$, $a > 0$ and

$$g(x) = ax, a > 0$$

are shown in the diagram above.

The graphs intersect when $x = 0$ and when $x = m$. ($m \neq 0$)

i) Show that $m^2 = 2a$. 2

ii) If the area of the shaded region is 64 square units, find the value of a and m . 2

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

SOLUTIONS

1. D

2. A

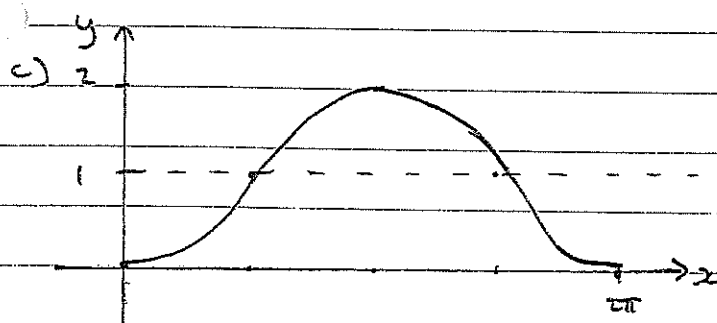
3. B

4. C

5. A

6. a) 0.061

b) $-\frac{\sqrt{3}}{2}$



d) $\int_0^{\frac{\pi}{6}} \sin 2x \, dx$
 $= \left[-\frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{6}}$
 $= \left(-\frac{1}{2} \cos \frac{\pi}{3} \right) - \left(-\frac{1}{2} \cos 0 \right)$

$$= -\frac{1}{4} + \frac{1}{2}$$

$$= \frac{1}{4}$$

e) $V = \pi \int_{-1}^0 x^2 \, dy$

$$= \pi \int_{-1}^0 y+1 \, dy$$

$$= \pi \left[\frac{1}{2} y^2 + y \right]_{-1}^0$$

$$= \pi \left((0) - \left(\frac{1}{2} - 1 \right) \right)$$

$$= \frac{\pi}{2} \text{ cu. units}$$

7. a) $\tan \theta = \frac{1}{\sqrt{3}}$
 $\theta = \frac{\pi}{6}, \frac{7\pi}{6}$

b) i) $3 \sec^2 3x - \sin x$

ii) $e^{2x} + 2(x+2)e^{2x}$
 $= e^{2x}(2x+5)$

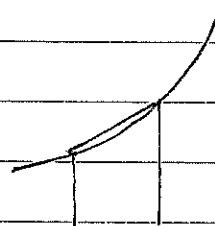
c) i)

x	1	3	5	7
$f(x)$	$\frac{1}{5}$	$\frac{9}{7}$	$\frac{25}{9}$	$\frac{49}{11}$

$$\therefore \int_1^7 f(x) \, dx \approx \frac{2}{2} \left[\frac{1}{5} + \frac{49}{11} + 2 \left(\frac{9}{7} + \frac{25}{9} \right) \right]$$

$$= 12.8$$

ii) curve is increasing and concave up



$\therefore$ each trapezium larger than actual area

$\therefore$ overestimate

8.

a. $60^\circ = \frac{\pi}{3}$

i) $l = r\theta$
 $= 10 \times \frac{\pi}{3}$
 $= \frac{10\pi}{3} \text{ m}$

ii) $\tan 60^\circ = \frac{BC}{10}$
 $BC = 10\sqrt{3}$

Area = $\Delta ABC - \text{sector}$

$$= \frac{1}{2} \times 10\sqrt{3} \times 10 - \frac{1}{2} \times 10^2 \times \frac{\pi}{3}$$

$$= 50\sqrt{3} - \frac{50\pi}{3} \text{ m}^2$$

b. $V = 2\pi \int_0^2 4^2 - (x^2)^2 dx$

$$= 2\pi \left[16x - \frac{1}{5}x^5 \right]_0^2$$

$$= 2\pi \left[32 - \frac{32}{5} \right]$$

$$= \frac{256\pi}{5} \text{ cu units}$$

c. $y = e^{4x} + x$

$$y' = 4e^{4x} + 1$$

when $x=0$

$$y' = 4e^0 + 1$$

$$= 5$$

$$y = e^0 + 0$$

$$= 1$$

$\therefore m=5$ passing thru' $(0,1)$

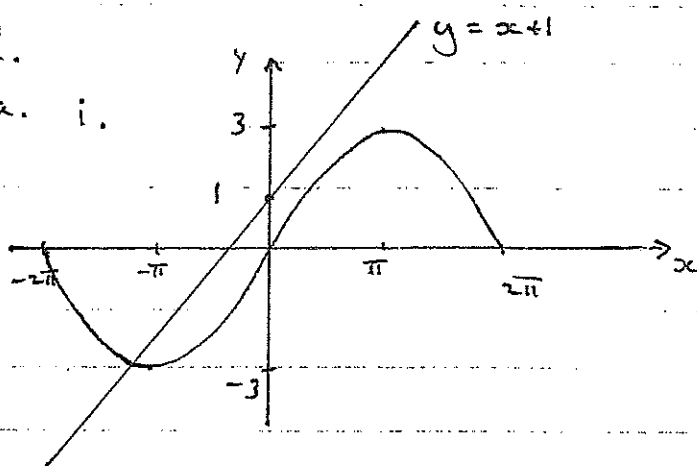
$$y - y_1 = m(x - x_1)$$

$$y - 1 = 5(x - 0)$$

$$5x - y + 1 = 0$$

9.

a. i.



ii. on diagram $y = x + 1$

iii. 1 solution

b. i. $y = e^x \sin x$

$$\therefore \frac{dy}{dx} = e^x \sin x + e^x \cos x$$

$$\therefore \frac{d^2y}{dx^2} = e^x \sin x + e^x \cos x + e^x \cos x - e^x \sin x$$

$$= 2e^x \cos x$$

ii. $\int e^x \cos x dx$

$$= \frac{1}{2} (e^x \sin x + e^x \cos x) + C$$

c. $y = 3e^{kx}$
 $y' = 3ke^{kx}$
 $y'' = 3k^2 e^{kx}$

$$\therefore y'' - 9y = 0$$

$$3k^2 e^{kx} - 27e^{kx} = 0$$

$$3e^{kx} (k^2 - 9) = 0$$

$$\therefore k = \pm 3$$

10.

a. i. $f(x) = \frac{e^{-x}}{x}$

$$f'(x) = \frac{-xe^{-x} - e^{-x}}{x^2}$$

$$= \frac{-e^{-x}(x+1)}{x^2}$$

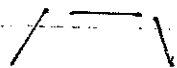
ii. st. pts when $f'(x) = 0$

$$\therefore x+1=0 \quad (e^{-x} \neq 0)$$

$$x = -1$$

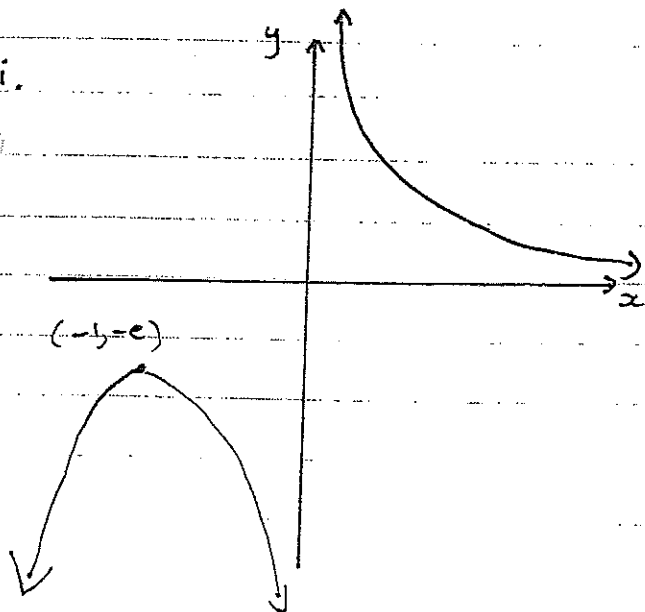
$$\therefore y = -e$$

test	x	-2	-1	$-\frac{1}{2}$
y'		ve	0	-ve



$\therefore$ max at $(-1, -e)$

iii.



b. i. $x^3 - ax = ax$

$$x^3 - 2ax = 0$$

$x=m$ satisfies this equation

$$\therefore m^3 - 2am = 0$$

$$m(m^2 - 2a) = 0$$

$$m^2 - 2a = 0 \quad (m \neq 0)$$

$$\therefore m^2 = 2a$$

ii. $\int_0^m ax - (x^3 - ax) dx = 64$

$$\int_0^m 2ax - x^3 dx = 64$$

$$\left[ax^2 - \frac{1}{4}x^4 \right]_0^m = 64$$

$$(am^2 - \frac{1}{4}m^4) - (0) = 64$$

but $m^2 = 2a$

$$a(2a) - \frac{1}{4}(2a)^2 = 64$$

$$a^2 = 64$$

$$a = 8 \quad (a > 0)$$

$$\therefore m = 4$$